

Question Paper

Exam Date & Time: 27-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
SEVENTH SEMESTER B.TECH. DEGREE EXAMINATIONS - NOVEMBER 2025

Ethical hacking and cyber security [IT_4024]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) Analyze how ethical hacking differs from malicious hacking and how intent defines legality. (5)
A) Course Outcome: "CO1" / Cognitive Level: "Analysis"
B) Analyze the effectiveness of port-scanning tools (Nmap, Unicorn Scan) in identifying open services in a network. (5)
Course Outcome: "CO2" / Cognitive Level: "Analysis"
- 2) Analyze how null sessions and weak authentication contribute to privilege escalation attacks in Windows systems. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Analyze how tools like DumpSec and Hyena simplify enumeration and risk discovery in enterprise networks. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 3) Examine the impact of delayed patching on zero-day vulnerability exploitation. (5)
A) Course Outcome: "CO3" / Cognitive Level: "Analysis"
B) Analyze how poor database connectivity security leads to injection and credential-leak issues in web servers. (5)
Course Outcome: "CO3" / Cognitive Level: "Analysis"
- 4) Analyze the importance of penetration-testing tools such as Burp Suite and OWASP ZAP in web application security assessment. (5)
A) Course Outcome: "CO2" / Cognitive Level: "Analysis"
B) Examine and analyze how phishing and spyware attacks compromise user credentials and analyze defense strategies. (5)
Course Outcome: "CO4" / Cognitive Level: "Analysis"
- 5) Examine and analyze the role of cyber forensics in tracing digital evidence in cybercrime investigations. (5)

A)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

B) Analyze the ethical and legal implications of hacking for defensive purposes under Indian cyber law. (5)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4024:Ethical hacking and cyber security

#	Course Outcome	Description	Marks
1	1	CO 1	5
2	2	CO 2	15
3	3	CO 3	15
4	4	CO 4	5
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025 IT_4024:Ethical hacking and cyber security

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	0	0(0%)
4	Analysis	10	50(100%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
Total		10	50 (100%)